

Technical Description:

Airpods (2nd Generation)

Author: Alexis Adkins

Intended Audience: College Students and Young Adults

Date: February 25, 2024

Content

The target audience is informed consumers looking into which wireless headphones would fit them best. More specifically, this audience consists of college students and young adults between the ages of 18–30 who regularly use wireless earbuds for music, calls, and media consumption. The purpose of this technical description is to ensure that informed consumers are provided with important information about AirPods (2nd Generation).

As informed consumers, they are likely to have knowledge of the features and options of wireless earbuds. While they are likely to be well-informed, they generally do not have specialized technical knowledge. In this description, information will be presented in layperson terms that match the general technical familiarity of college students and young adults who are comfortable with everyday technology but may not understand advanced specifications.

They may need this description to compare it to other products. This description will help informed consumers to compare the specs of AirPods (2nd Generation) to other wireless earbuds on the market.

Introduction

AirPods (2nd Generation) were released on March 20, 2019, by Apple Inc. These wireless Bluetooth earbuds are designed primarily for use with Apple devices, such as iPhones, iPads, and MacBooks.



Compared to the original version, AirPods (2nd Generation) offer improved audio and voice quality, longer talk time, and support for voice-activated Siri (Wikipedia, 2024). Their main purpose is to provide a convenient, wireless way for users to listen to audio content throughout the day, whether they are studying, commuting, or relaxing. (See *Figure A*)

Figure A – AirPods (2nd Generation)

Details

External-

(1) Size/Dimensions:

The earbud of a set of AirPods (2nd Generation) weighs .14 ounces each. Each earbud has a height of 1.59 inches and a width of 0.65 inches. The depth of each earbud is 0.71 inches. The wireless charging case weighs 1.35 oz. The height is 2.11 inches, and the width is 1.74 inches. The depth is .84 inches (*See Figure B*) (“Apple,” n.d.).

(2) Shape:

Every version of AirPods features a distinctive earbud-and-stem design with AirPods (2nd Generation) offering an elongated stem that doesn’t seal to your ear. The AirPods (2nd Generation) stand out with a non-intrusive design that avoids creating a seal with the ear canal while providing a comfortable fit. The case is shaped like a rounded square-shaped with a flip-top lid (*See Figure B*) (Gideon, 2021).

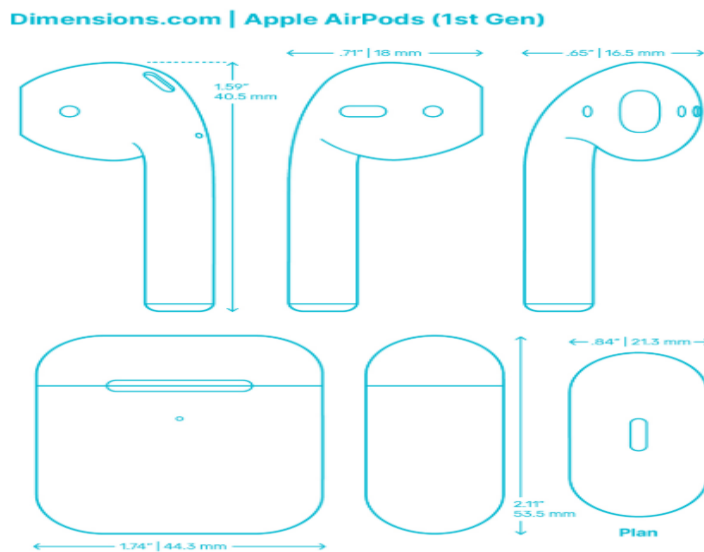


Figure B – Airpods (2nd Generation) Dimensions and Shape

(3) Color:

AirPods (2nd Generation) are available in one color: white.

Internal-

(1) Components and Materials:

AirPods (2nd Generation) feature four distinct sensors: dual beamforming microphones, dual optical sensors, a motion-detecting accelerometer, and a speech-detecting accelerometer. These sensors have multiple functions, including enhancing audio quality, determining when the AirPods are actively in use, monitoring AirPod movement, and recognizing when the user is speaking into the device (“Airpods,” n.d.). It also has an

Apple H1 headphone chip. This chip guides with noise cancellation, in-ear detection, and device switching (Triggs, 2023).

(2) Features:

The AirPods use touch-based controls located on the head of the stem (*See Figure C*). These controls allow users to play or pause audio, skip forward or backward between tracks, and activate Siri.

The sensors enable these controls by detecting taps and movement. For example, users can double-tap to skip a song or access Siri without needing to touch their phone (“Pause,” 2025). This is especially useful for college students who are often multitasking or on the go.



Figure C – Touch Sensors

Functionality

These parts work together because the sensors in the AirPods enable the Touch control to function using motion detection. You can use the Touch control on the Airpod's stem to play or pause audio and mute or unmute yourself during a call. Additionally, you can skip forward by double-tapping the Touch control or go back by triple-tapping. When adjusting the volume, just place your thumb on the Touch control and swipe up or down. Thanks to voice detection, Siri can also be activated and utilized with the AirPods (“Pause,” 2025).

Conclusion

This technical description is intended for college students ages 18 to 24 who are considering AirPods (2nd Generation) as their wireless earbuds. Released in March 2019, these earbuds offer improved audio quality, voice support through Siri, and user-friendly controls.

The external details cover dimensions and design, including a portable charging case. Internally, they feature four sensors and an Apple H1 chip for enhanced service, including in-ear detection.

The description is to inform consumers when comparing the AirPods (2nd Generation) with other wireless earbuds in a user- friendly manner. AirPods (2nd Generation) can enrich the lives of consumers by providing high-quality audio that enhances their enjoyment of music, podcasts, calls, and other media while being integrated into the user's lifestyle.

References:

- “Airpods.” (n.d.) *AirPods 2 - Tech Specs - Apple Support*. Apple Support. <https://support.apple.com/en-us/111856>
- Amazon.com: Apple AirPods (2nd Gen) Wireless Ear Buds + Lightning Charging Case + Lightning to USB Cable + 1 year Limited Warranty : Electronics.* (n.d.). <https://www.amazon.com/Apple-Wireless-Lightning-Charging-Warranty/dp/B0DDX4BT2Z>
- “Apple.” (n.d.). AirPods (2nd generation) - Technical Specifications. <https://www.apple.com/airpods-2nd-generation/specs/>
- Dimensions.com. (n.d.). *Apple AirPods (2nd gen) dimensions & drawings | Dimensions.com.* <https://www.dimensions.com/element/apple-airpods-2nd-gen-2019>
- Gideon, T. (2021, October 25). Apple Earbuds Do Battle: AirPods (3rd Gen) vs. AirPods (2nd Gen) vs. AirPods Pro. PCMAG. <https://www.pcmag.com/news/apple-earbuds-do-battle-airpods-3rd-gen-vs-airpods-2nd-gen-vs-airpods-pro>
- “Pause.” (2025, September 2025). *Pause, skip, and adjust volume with your AirPods and AirPods Pro - Apple Support*. Apple Support. <https://support.apple.com/en-us/102628>
- Triggs, R. (2023, March 16). Explained: What do the Apple H1 and H2 chips do? Android Authority. <https://www.androidauthority.com/apple-h1-h2-chips-979713/>
- Wikipedia contributors. (2024, January 31). AirPods. Wikipedia. <https://en.wikipedia.org/wiki/AirPods>